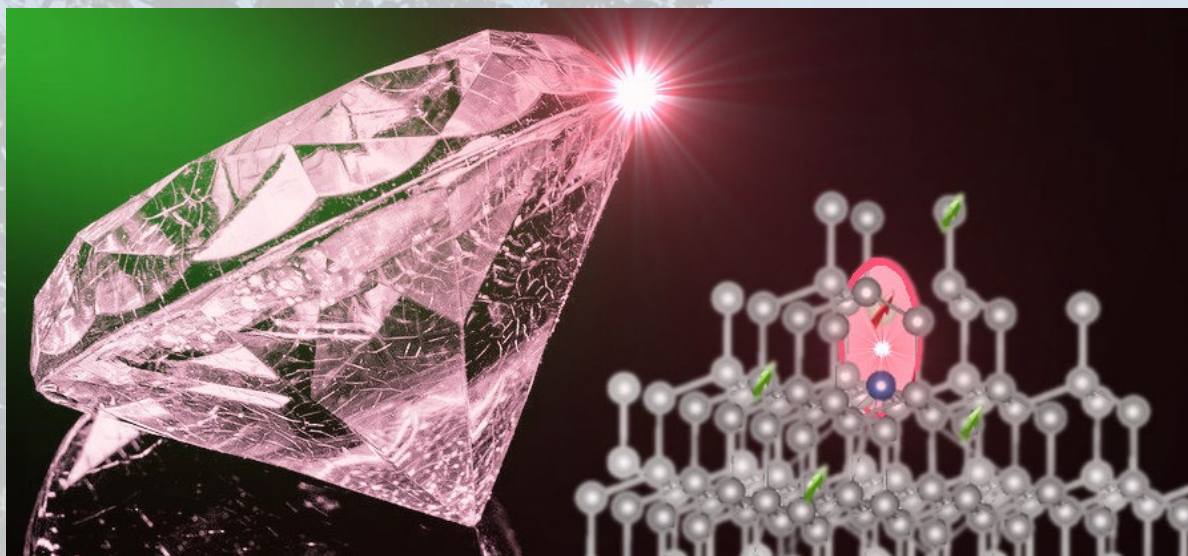




Dr. Walter Hahn, Fraunhofer IAF

Quantum computing at Fraunhofer IAF

Contents of the talk

Diamond quantum computing und algorithms research @ Fraunhofer IAF

1. Fraunhofer Society and Fraunhofer IAF at a glance
2. Overview of the Business Unit Quantum Systems am Fraunhofer IAF
3. Introduction to diamond quantum technologies
4. Quantum computing hardware @ Fraunhofer IAF
5. Quantum computing algorithms @ Fraunhofer IAF
6. Summary and Outlook

Fraunhofer Society

At a glance

Applied research focusing on key future-relevant technologies and the commercialization of findings in business and industry.



> 30,000
employees



76 institutes
and research units

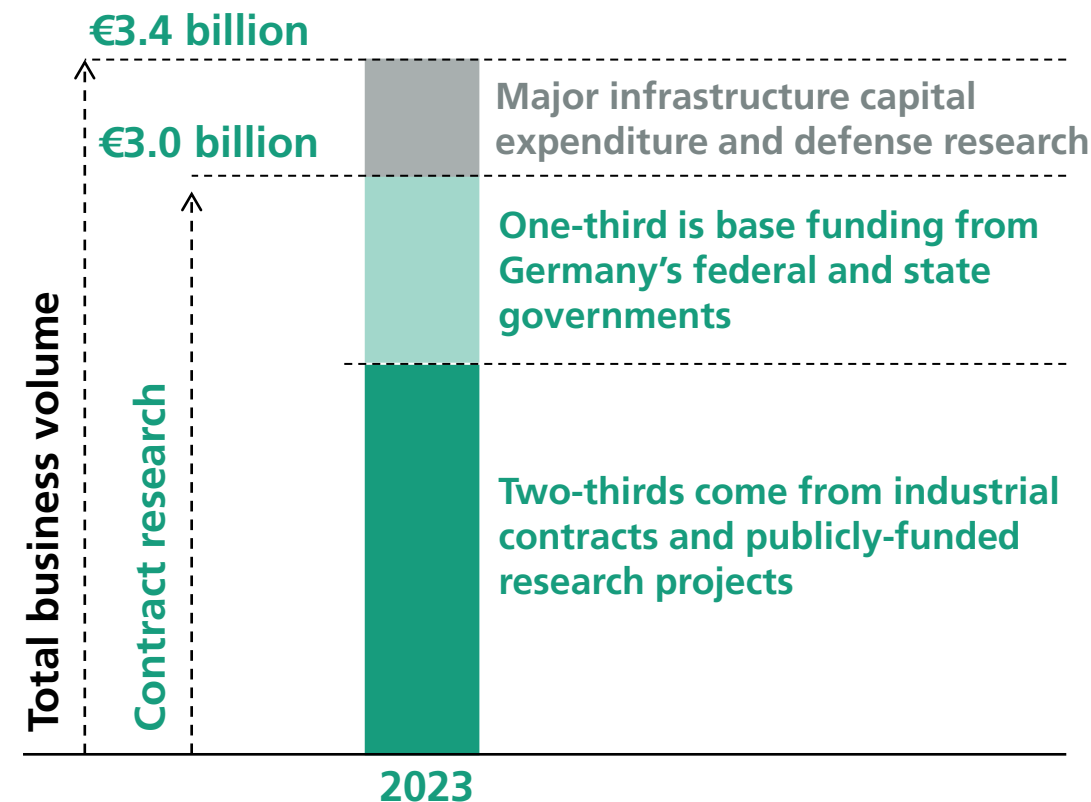


€3.4 billion business volume
€3.0 billion contract research



Fraunhofer Society

At a glance



Fraunhofer Competence Network Quantum Computing

In order to advance applied research on quantum computing, the Fraunhofer-Gesellschaft has established a national network based on competence centers: The Fraunhofer Competence Network Quantum Computing

One of these competence centers is the Competence Center Quantum Computing Baden-Württemberg, headed by Fraunhofer IAF and IAO.

Fraunhofer-Kompetenznetzwerk Quantencomputing

- Mitglied
- Gast

Fraunhofer CML

Optimierung für Logistik und Schifffahrt

Fraunhofer FOKUS

Plattformen, Zertifizierung, Sicherheit

Fraunhofer IIS-EAS, IAIS-EST, IOSB-AST, IWU

Optimierung, Simulation

Fraunhofer IMS

Halbleiterelektronik und Sensoren

Fraunhofer IOF

Photonische Plattformen, optische Systemtechnik und Komponenten

Fraunhofer ILT

Entwicklung und Optimierung von Hardware
Photonische (Komponenten und) Systemtechnik

Fraunhofer SCAI, IAIS

Machine Learning auf Quantencomputern, Quantum KI

Fraunhofer ITWM

Quanten-HPC, Algorithmen

Fraunhofer IGD, SIT

Algorithmen, heterogenes Rechnen

Fraunhofer IAO, IPA

Software-Engineering, Hybride Anwendungen, Machine Learning

Fraunhofer IIS

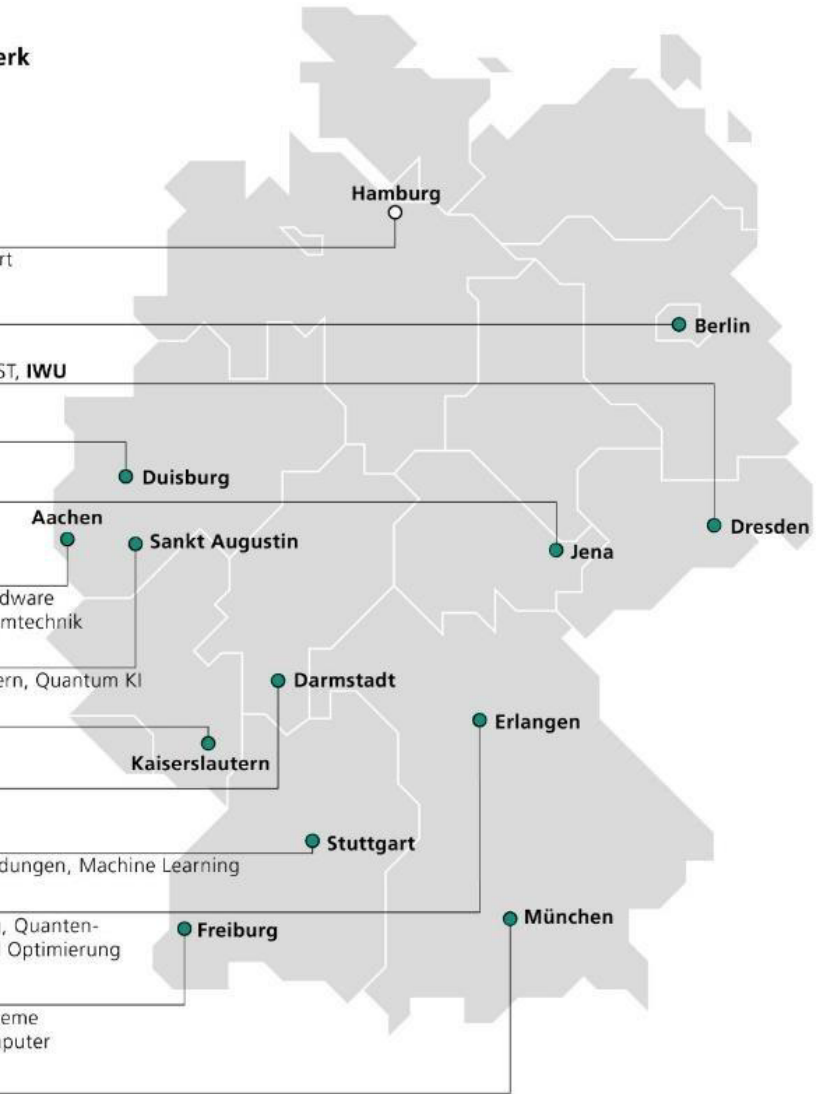
Elektronik zur Messung und Steuerung, Quanten-
algorithmen für Machine Learning und Optimierung

Fraunhofer IAF, IWM

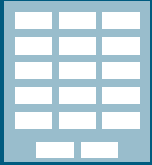
Quantenhardware, Hybride Rechensysteme
Materialmodellierung mit Quantencomputer

Fraunhofer AISEC, IKS

Sicherheit, Robustheit, Optimierung



Fraunhofer IAF at a Glance



1957
founded in Freiburg



45.2 Mio € p.a.
total budget



289
employees



Institute Management
Dr. Patricie Merkert
Prof. Dr. Rüdiger Quay



1000 m²
clean room
450 m²
MOCVD machine hall



ISO 9001

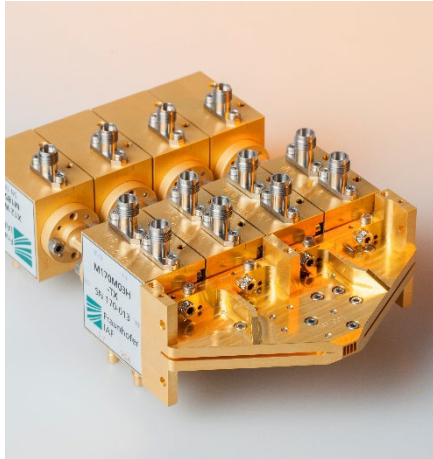
Certified
Quality Management System

www.tuv-sud.com/ms-cert



The Fraunhofer Institute for Applied Solid-State Physics (IAF) at a Glance

The five Business Units



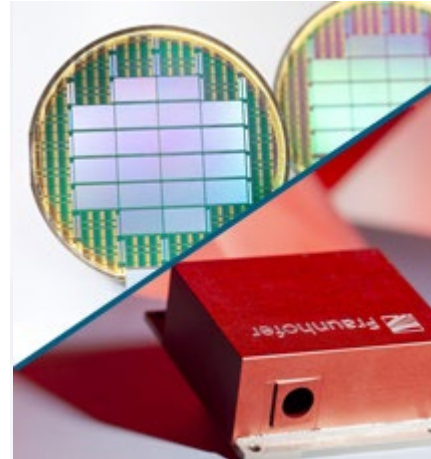
High-Frequency Electronics

Ultra high-frequency circuits for radar, communication, satellites and quantum computing



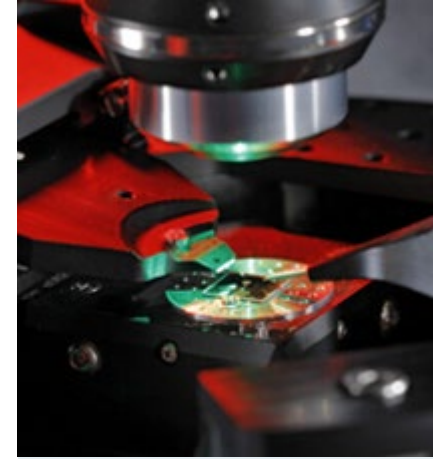
Power Electronics

Power electronics for mobile communication (incl. 5G & 6G), radar and energy conversion



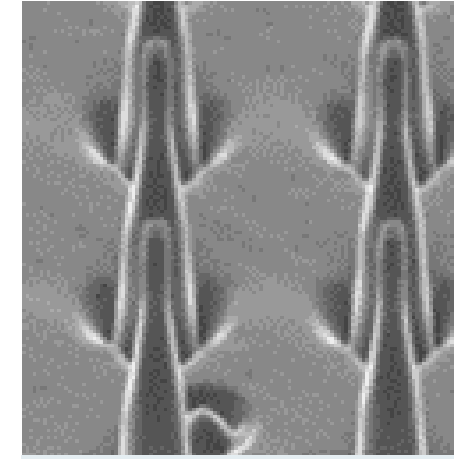
Optoelectronics

Detectors with high spatial resolution in the infrared and UV range and Infrared semiconductor lasers



Quantum Components

Quantum sensor technology in many variants and tuning with qubits



Quantum Systems

Solutions for quantum systems from qubits and quantum computers to quantum software

The Fraunhofer Institute for Applied Solid-State Physics (IAF) at a Glance

The five Business Units



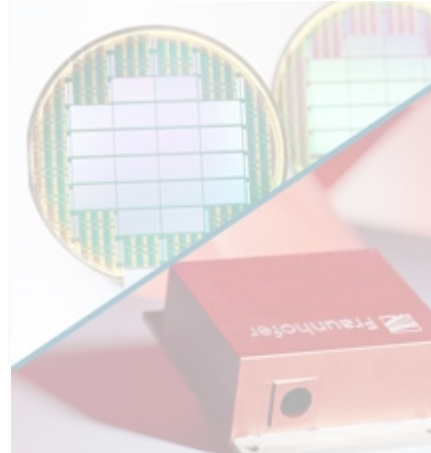
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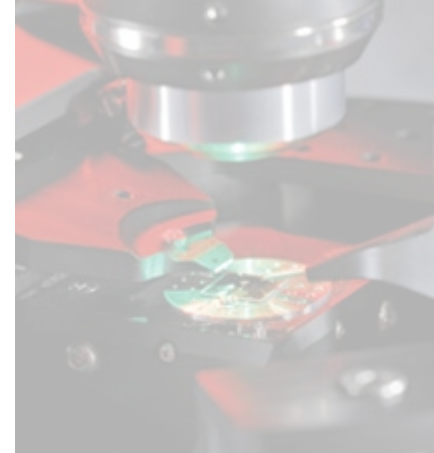
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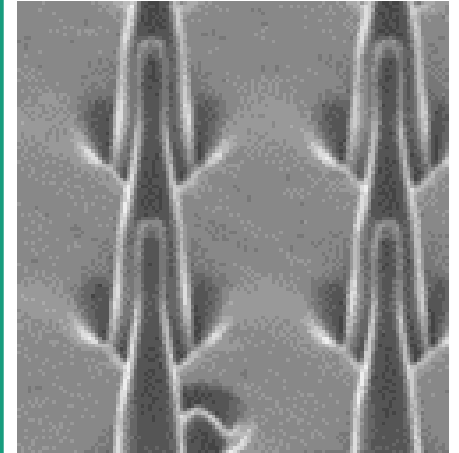
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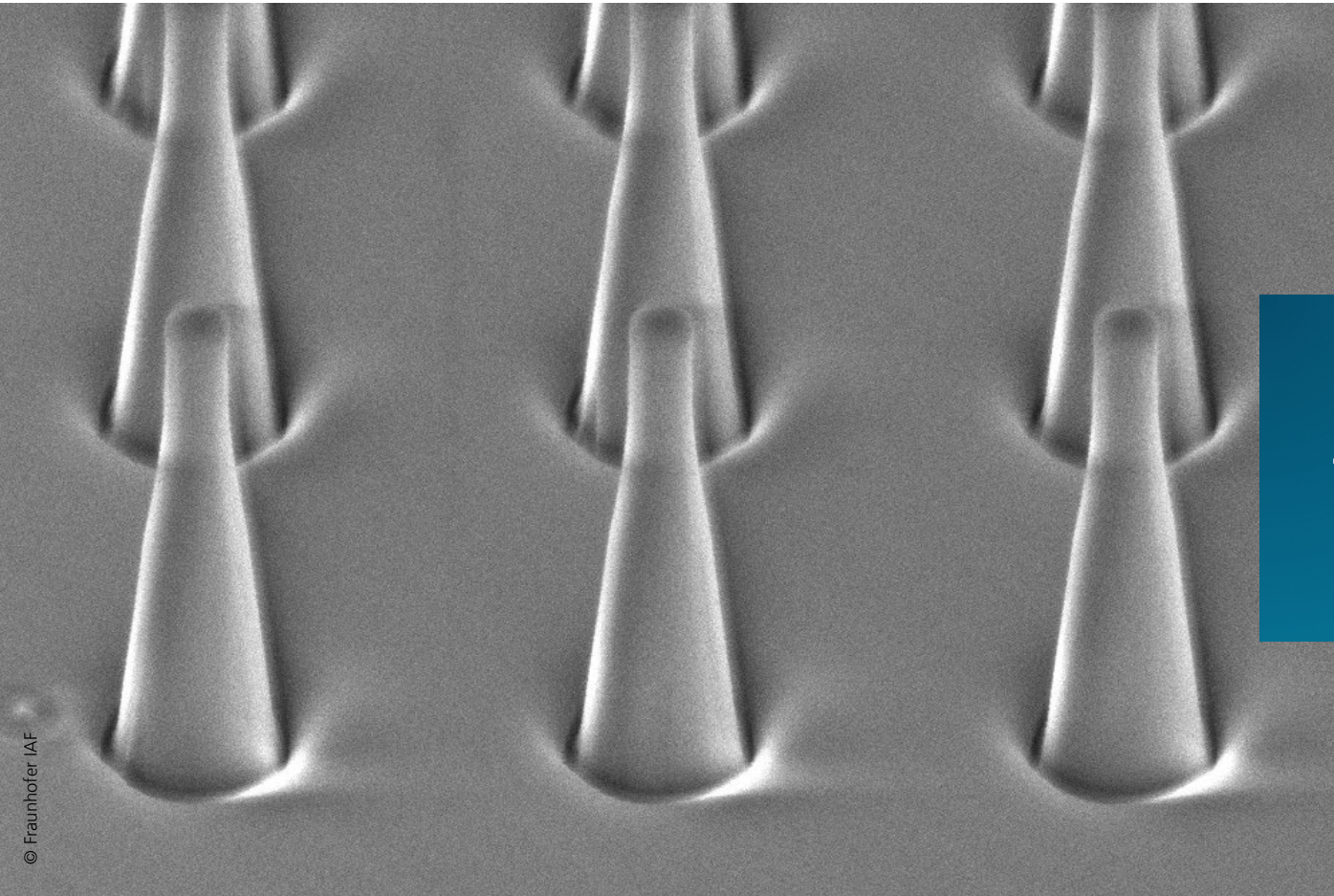


Quantum Systems

Solutions for quantum systems from qubits and quantum computers to quantum software

Quantum systems at Fraunhofer IAF

BU Leader: Dr. Florentin Reiter



Research on quantum systems –
from qubits to quantum algorithms

Quantum systems research at Fraunhofer IAF

BU Leader: Dr. Florentin Reiter

1

Material synthesis for diamond-based quantum technologies

2

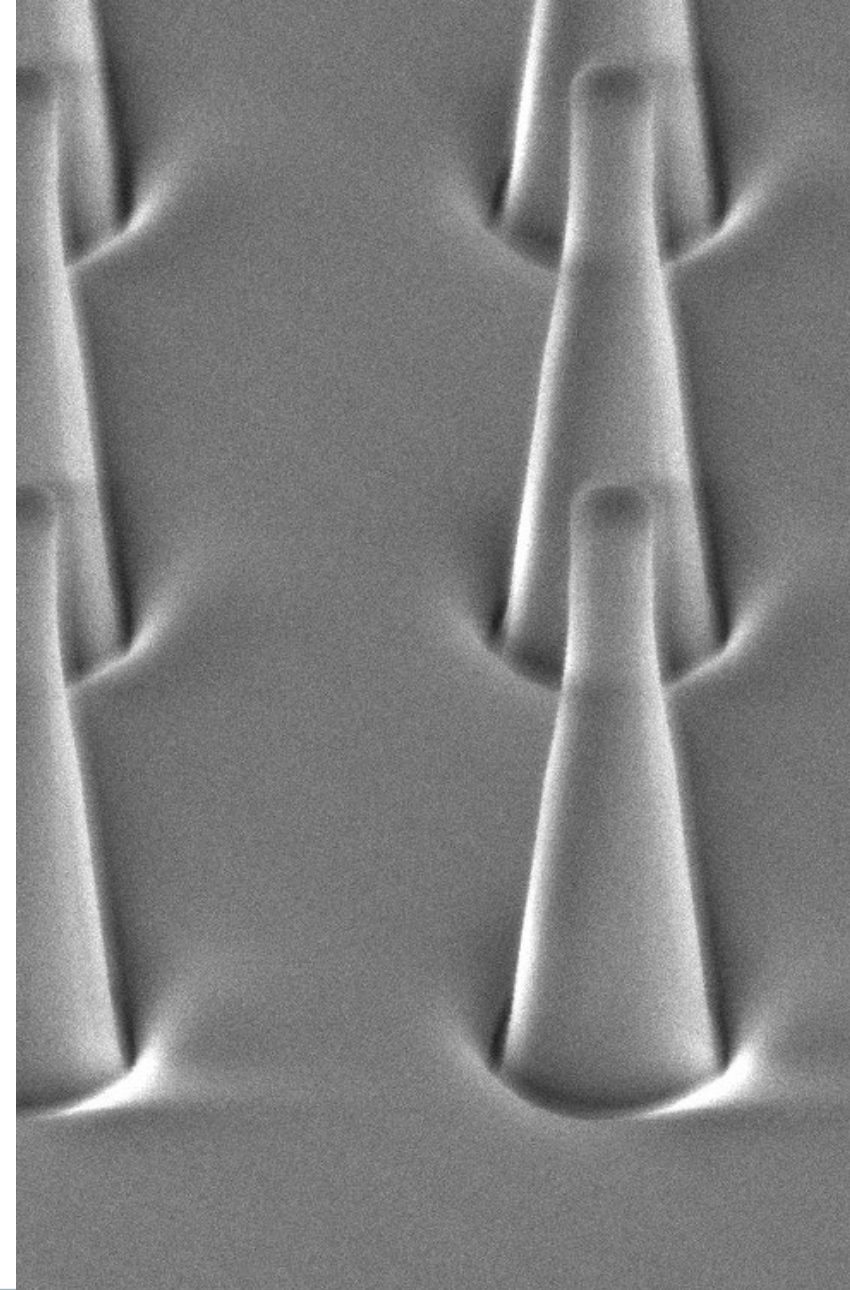
Development of quantum computer hardware based on spin qubits / color centers in diamond

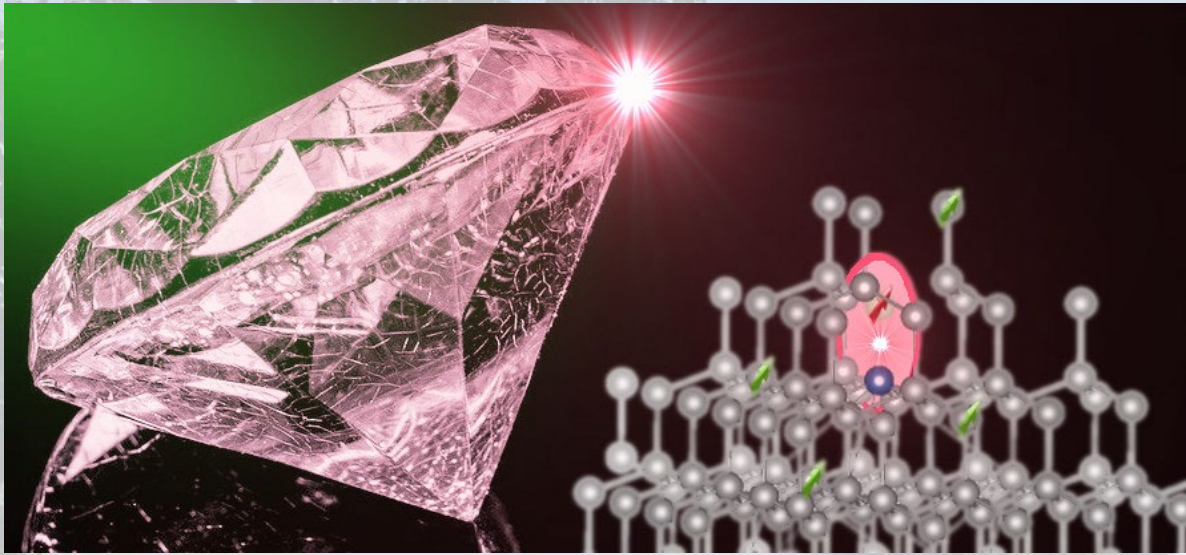
3

Characterization and simulation of quantum computing platforms

4

Design of hardware-optimized quantum algorithms for simulation, optimization, and machine learning





Quantum computing at Fraunhofer IAF

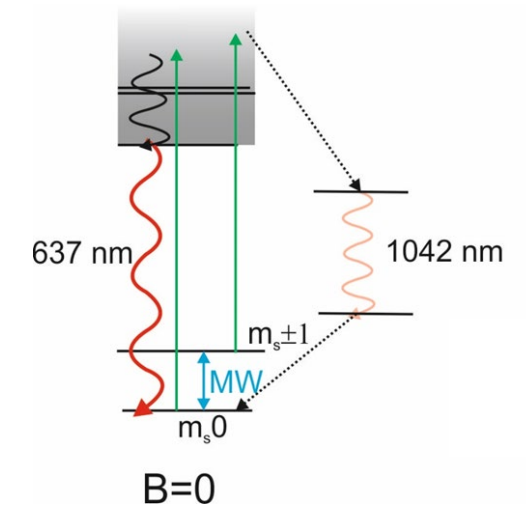
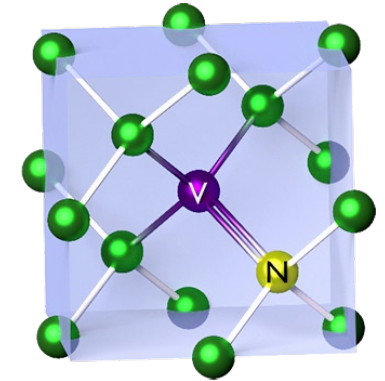
Diamond quantum technologies

Diamond quantum technologies

Motivation

The nitrogen-vacancy (NV) center in diamond

- Replace two carbon atoms of a diamond lattice by a **nitrogen** atom and a **vacancy** →
- Control and read-out of the electron spin through light and microwave fields
- Behaves like a trapped atom or ion – “nature’s own ion trap”
- **Unique properties and advantages:**
 - Well-defined fluorescence: green / red light

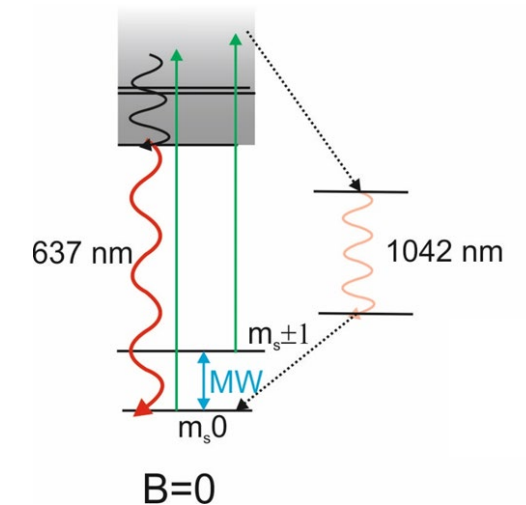
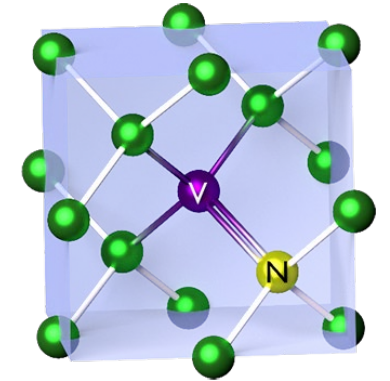


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 - Lifetimes naturally milliseconds, but up to **seconds** possible. ^{13}C nuclear spin up to **minutes**

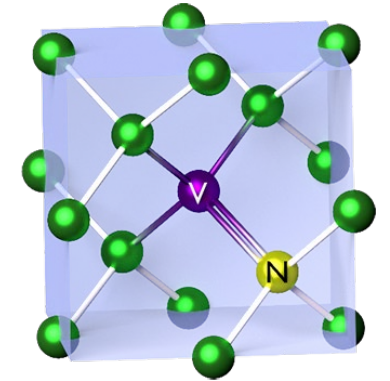


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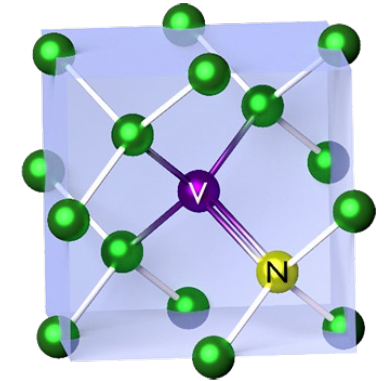


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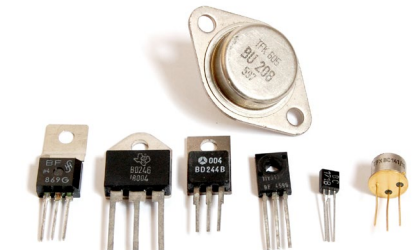
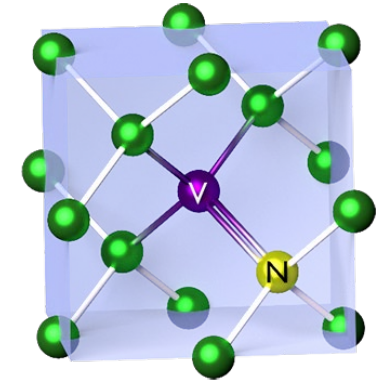


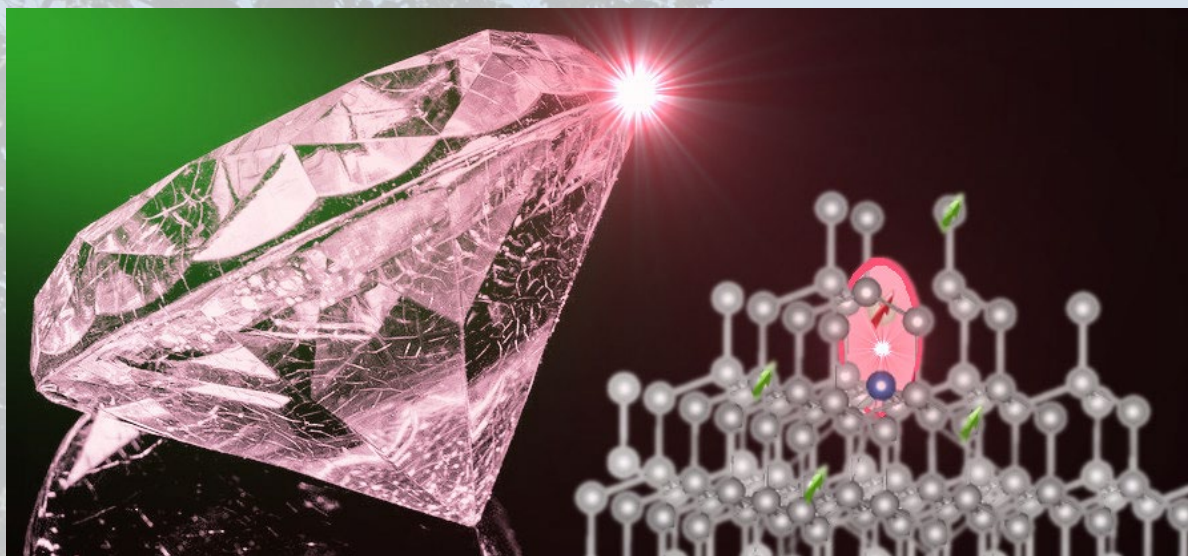
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 - Operation possible at **room temperature** → less effort / energy consumption
 - Atom-like but no trap required
 - Carbon, Nitrogen → cheap, abundant materials
- **Applications:**
 - NV center promises **mobile** and **cheaper** quantum devices
 - Quantum sensors, quantum communication, **quantum computers**



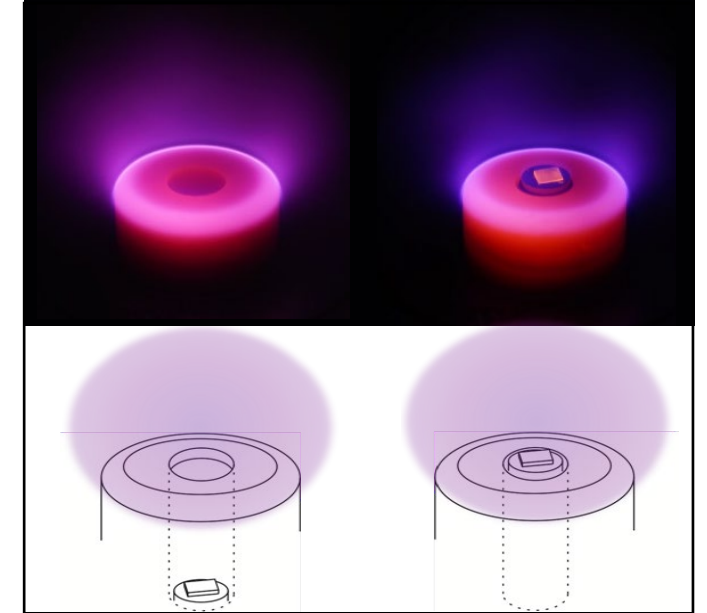
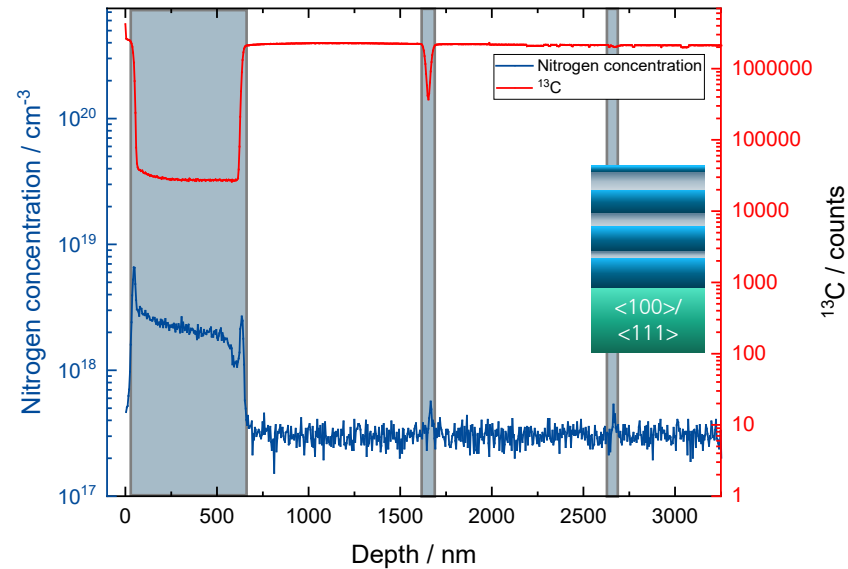


Quantum computing at Fraunhofer IAF

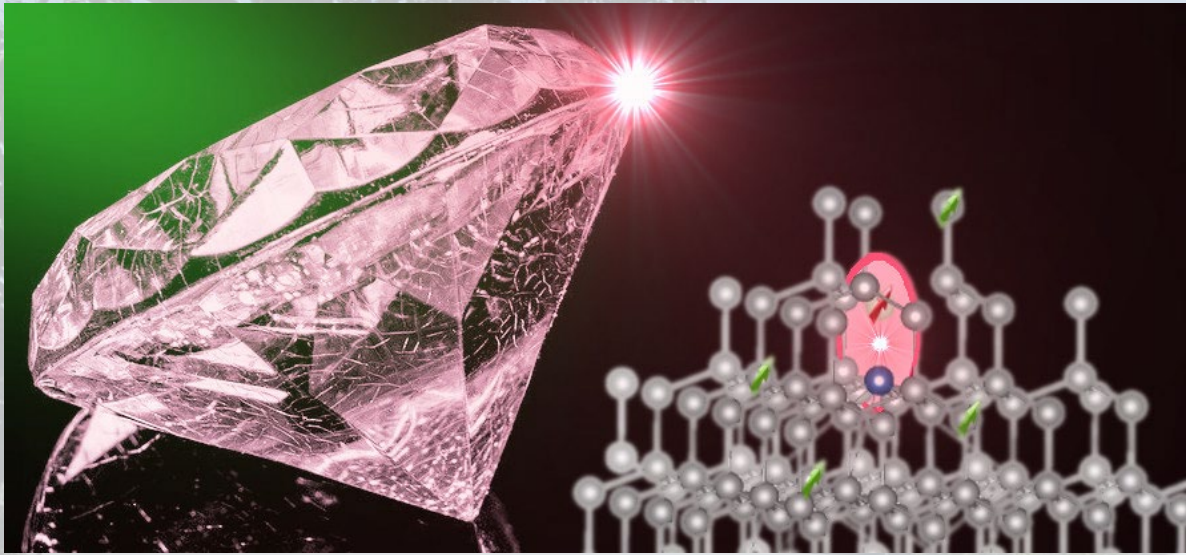
Creating the qubits

Diamond quantum computing – creating the qubits

Fabrication by doping



- Controlling isotopic ratios (¹²C/¹³C, ¹⁵N/¹⁴N)
- Doping with Nitrogen (other dopants possible)
- Capping of N-doped diamond layers with few nanometer thick layers
- Orientation of NV-centers via growth
- Statistical process → Difficult to obtain single NV centers



Quantum computing at Fraunhofer IAF

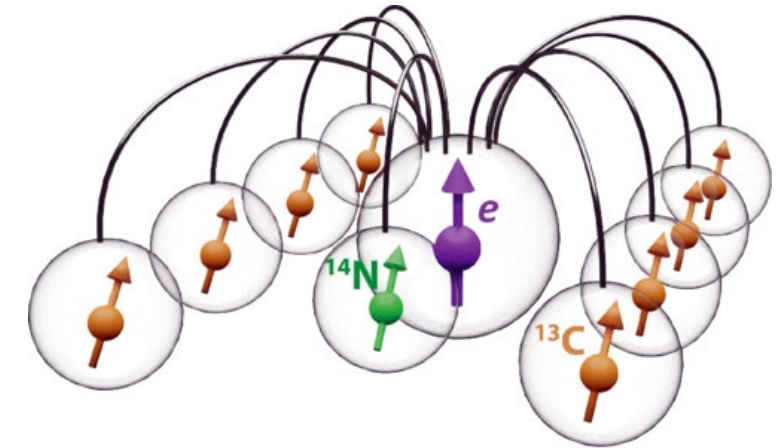
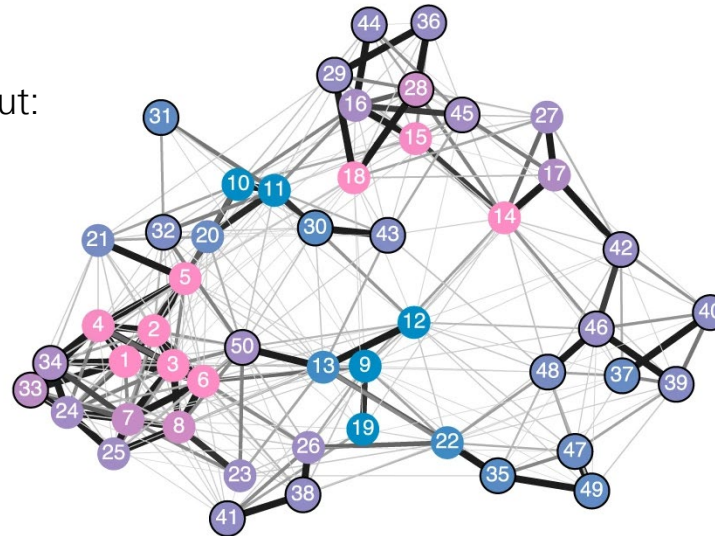
Diamond quantum computing –
physical paradigms

Diamond quantum computing

Clusters of qubits

- **Requirement:** have multiple well-defined controllable qubits and connect them (in a quantum-mechanical way)
- **NV-clusters:** Nuclear spins of Carbon-13 surrounding the electron spin can be used as qubits with high coherence times (up to several minutes)
Coupled through dipolar interaction
- Couplings of up to **50 spins** have been mapped out:
Taminiau group, TU Delft
- **Commercial products available from QuantumBrilliance, SaxonQ, XeedQ**
exhibit a number of well-controlled qubits, operating at room temperature

van de Stolpe, Nat. Commun. (2024)
DOI: 10.1038/s41467-024-46075-4

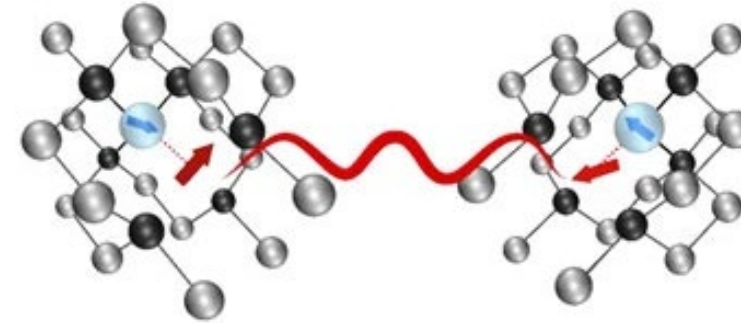


Bradley et al., Phys. Rev. X 2019,
DOI: 10.1103/PhysRevX.9.031045

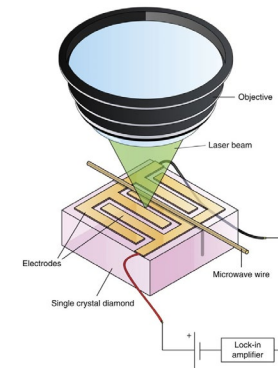
Diamond quantum computing

Clusters of qubits – coupled

- **Coupling several NV centers** (dipolar) positioned close to each other
 - For close-to-room-temperature operation: 10 nm and below
Nanoscale control on NV center position needed
Fabrication processes challenging
 - Readout (electronic) requires nanoscale precision
- Research effort under EU Horizons project **SPINUS**
12 partners from 7 countries, coordinated by Fraunhofer IAF
- **Goal:** demonstrate scalability > 10 qubits, towards 100 qubits

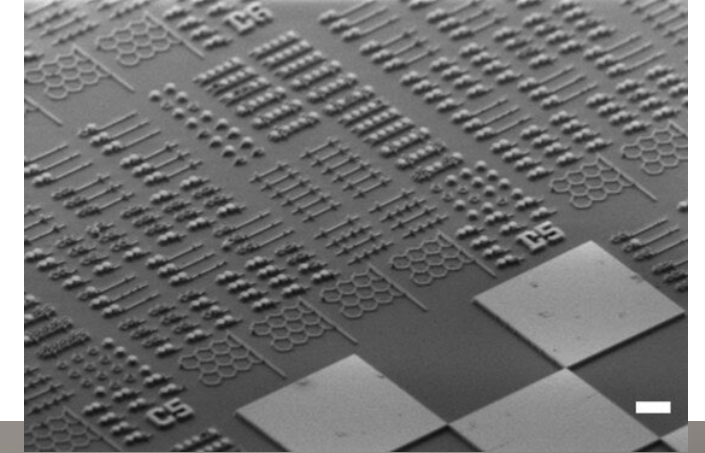


F. Dolde et al., Nat. Commun. (2014), DOI: 10.1038/ncomms4371
E. Bourgeois et al., Adv. Optical Mat. (2020), DOI: 10.1002/adom.201902132



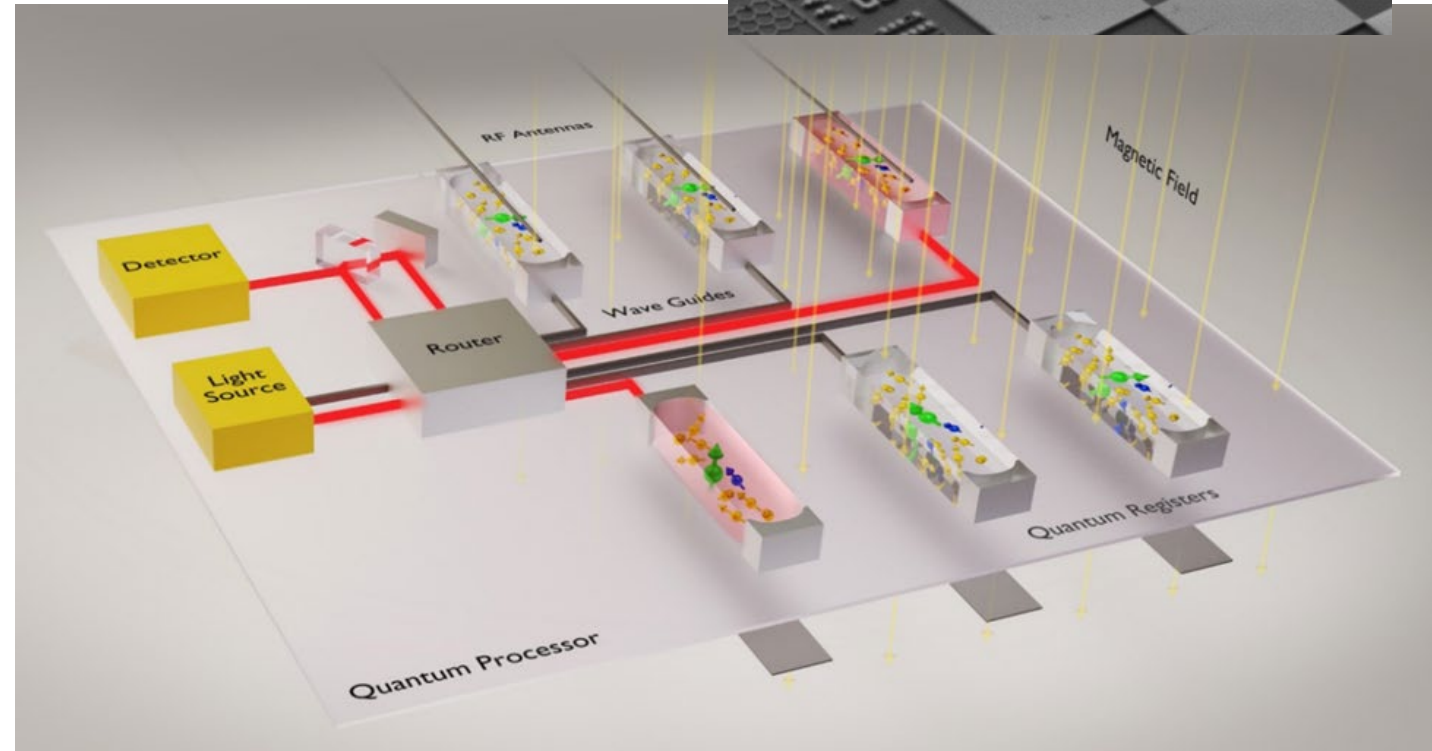
Diamond quantum computing

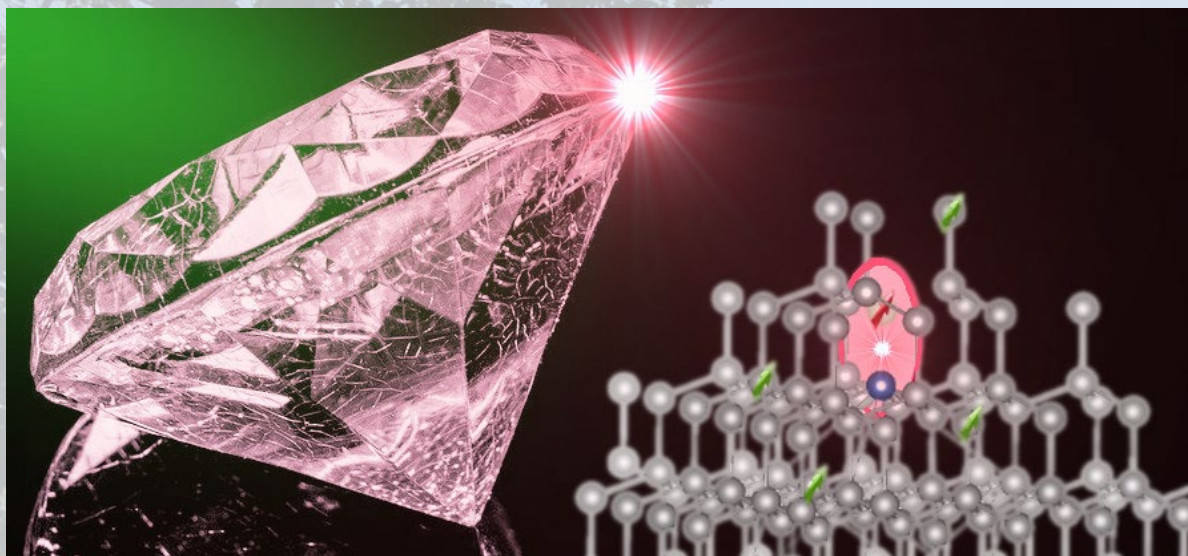
Hybrid spin-photon based quantum computer



Partners

- Scalable quantum computing
 - BMBF Project 01KT1901, 14 partners
 - Development of quantum control protocols
 - **Goal:** development of quantum registers
- Fraunhofer IAF (coordination)
 - Fraunhofer IISB
 - Forschungszentrum Jülich
 - Karlsruhe Institute of Technology
 - University of Konstanz
 - Heidelberg University
 - Technical University of Munich
 - University of Stuttgart
 - Ulm University
 - Diamond Materials GmbH
 - NVISION IMAGING Technologies GmbH
 - Qinu GmbH
 - Quantum Brilliance GmbH
 - Swabian Instruments GmbH
 - Qruise GmbH
 - 14 associated partners from science and industry





Quantum computing at Fraunhofer IAF

Diamond quantum computing at
Fraunhofer IAF

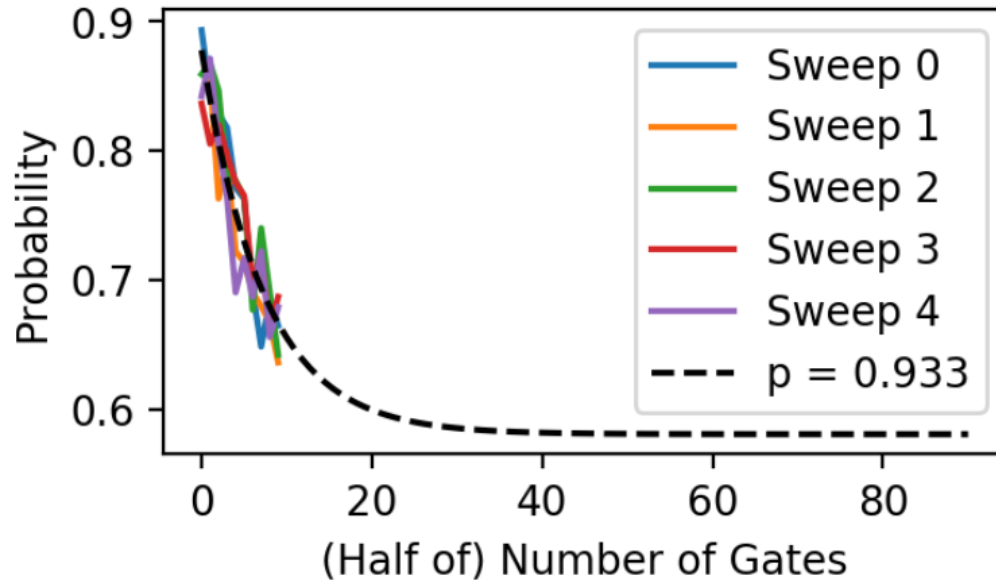
Diamond quantum computing at Fraunhofer IAF

Realization of two-qubit gates on a NV spin register @ IAF



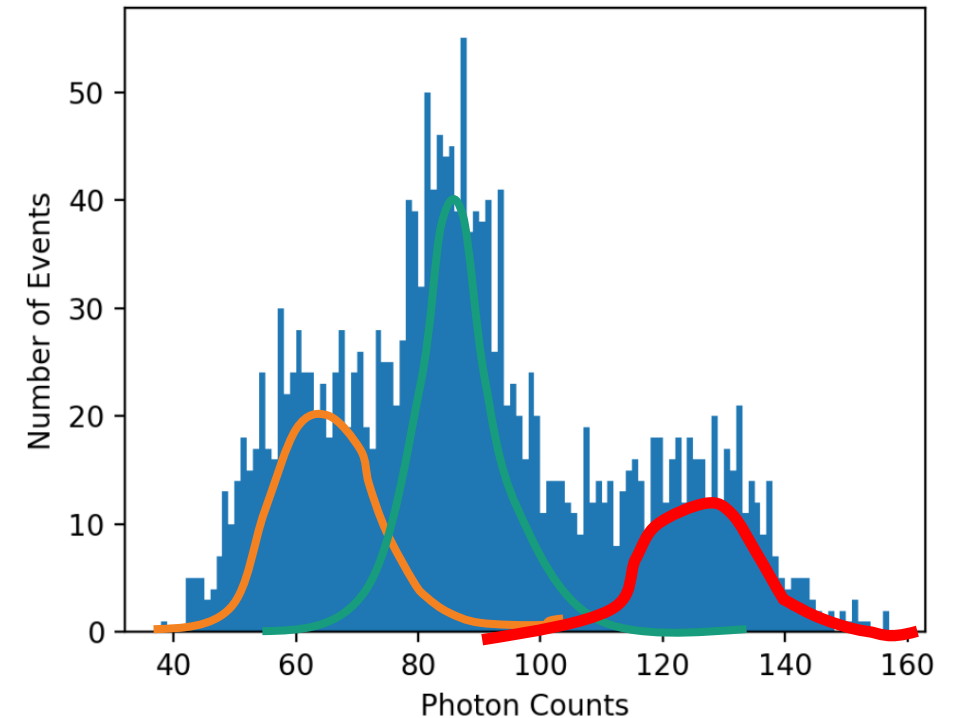
Realization of Two-Qubit Gates on a NV Spin Register @ IAF

A CNOT Between Two Nuclear Spins



Measured data over 4 sweeps to get an estimate of the fidelity

- The fit gives us a fidelity for each gate of $\mathcal{F} = p + \frac{(1-p)}{d} \approx 95\%$
- d : Dimensionality of $\rightarrow d = 2^2$
- Number of gates is the pair $(UU^\dagger)$



Single-shot measurement after the CNOT gate:

The SSR shown provides us with a ~90% probability that we are in the state we read out. In this case we read out: $|m_s = 0, m_I^N = \uparrow, m_I^{13C} = \uparrow\rangle$

SSR Literature: P. Neumann et al., Science **329**, 542-544 (2010)

Diamond quantum computing @ IAF

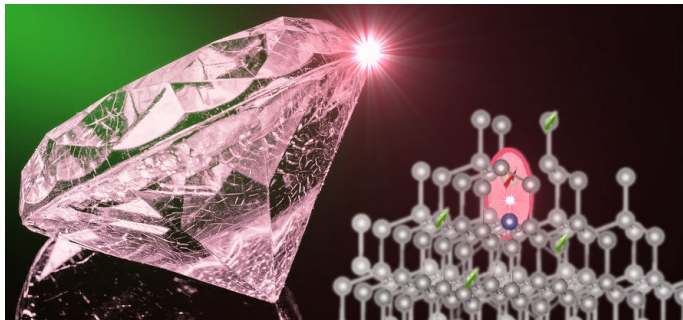
Short- and intermediate-term goals

Quantum
computing
systems

SQUTEC
system



Further
systems



HPC integration

(= high performance
computing system)

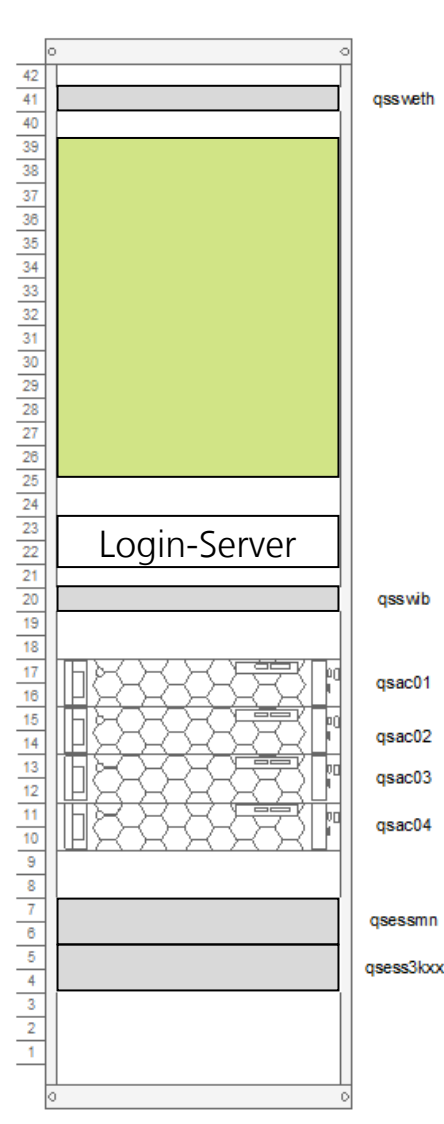
QPU + CPU



High performance computing system
(image indicative)

HPC Cluster at IAF

- NV-based Quantum computer
- Electron spin + 2 nuclear spins (2 qubits)
- Integrated into the HPC cluster already



- NVIDIA Grace Hopper Server
- Quantum Machines OPX1000 control system
- Fast PCI-express card connection with about 3μs roundtrip time
- Will be installed soon in the HPC cluster (quantum system in development)



Diamond quantum computing @ IAF

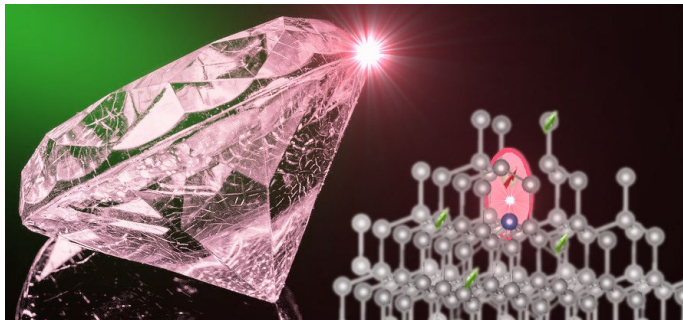
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Quantum
computing
systems

SQUTEC
system



Further
systems



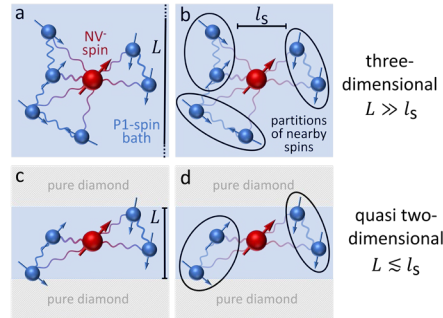
?



Quantum computing at Fraunhofer IAF

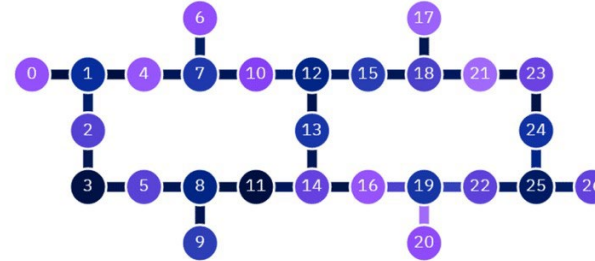
Quantum algorithms research – overview

Research group „Quantum Information“



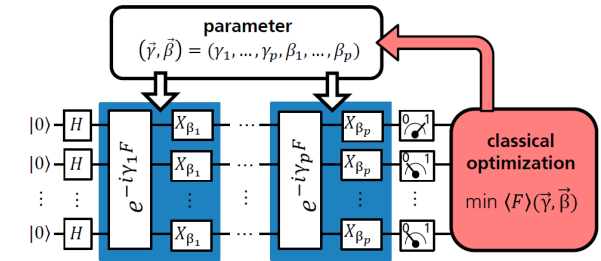
Quantum spin simulations

- Quantum spin dynamics for material development and quantum technologies
- pCCE method
Schätzle et al., PRB 110, L220302 (2024)
- Noise characterization



Quantum-Hardware analysis

- Benchmarking
- Error characterization (cross talk, improved noise models)
Ketterer et al., Phys. Rev. Applied 20, 034065 (2023)
- Error mitigation (inverted-circuit zero noise extrapolation)
König et al., PRA 110, 042625 (2024)

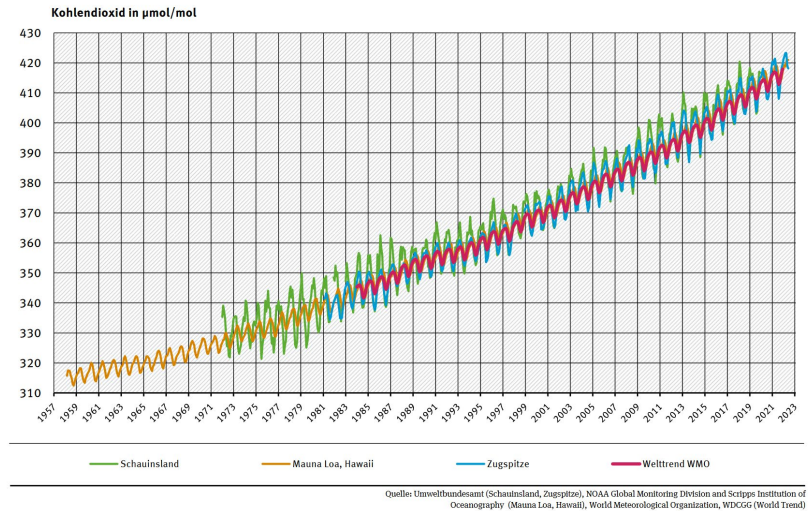


Quantum algorithms

- Combinatorial optimization: QAOA and LR-QAOA
Dehn et al., arXiv:2504.08577
- Quantum-chemistry calculations: VQE
Possel et al., arXiv:2505.19772
- Quantum Machine Learning

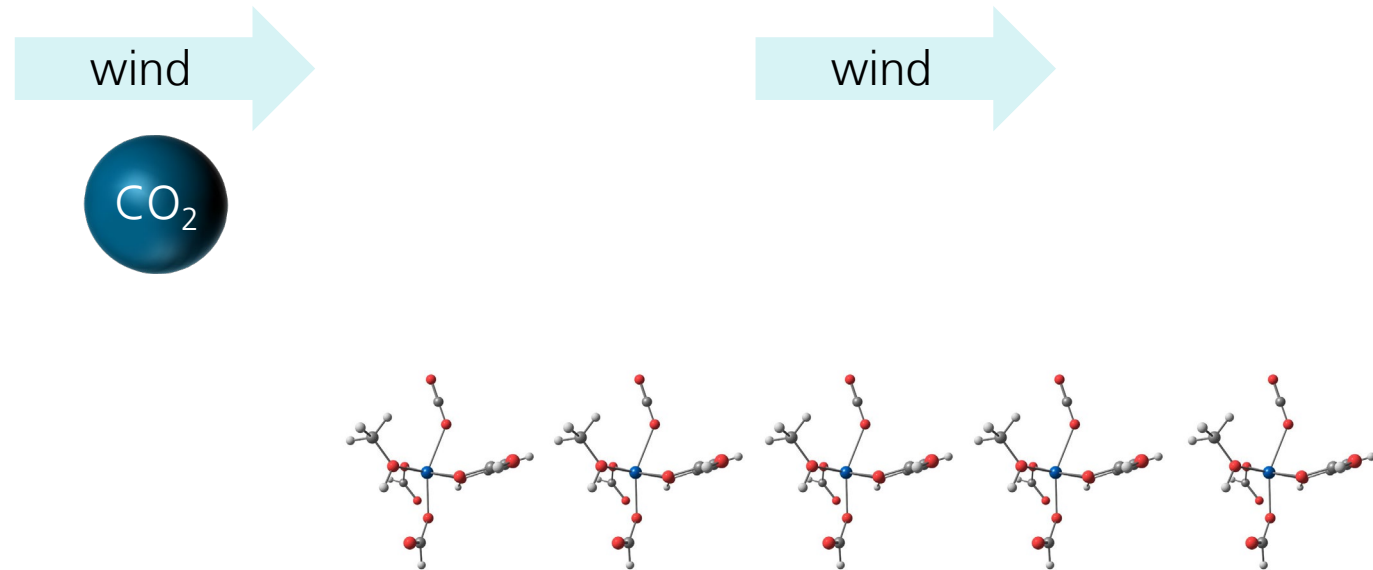
Problem: Reduction of the CO₂ concentration in the atmosphere

Kohlendioxid-Konzentration in der Atmosphäre (Monatsmittelwerte)



Reduction of the CO₂ concentration
in the atmosphere

Idea: direct CO₂ capture from the atmosphere



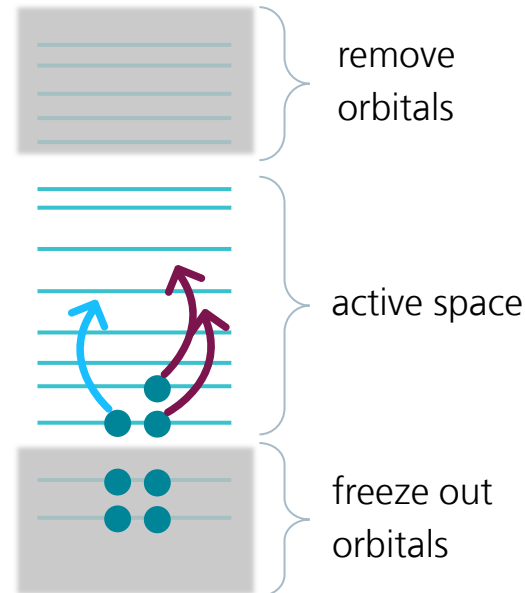
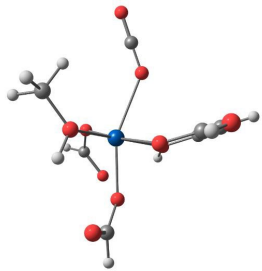
Long-term goal:

Simulation of the CO₂ capture by Metal-Organic Frameworks
on quantum computers

Quantum Chemistry with the Variational Quantum Eigensolver (VQE)

Main steps for using VQE:

- Determine the molecular orbitals with classical methods



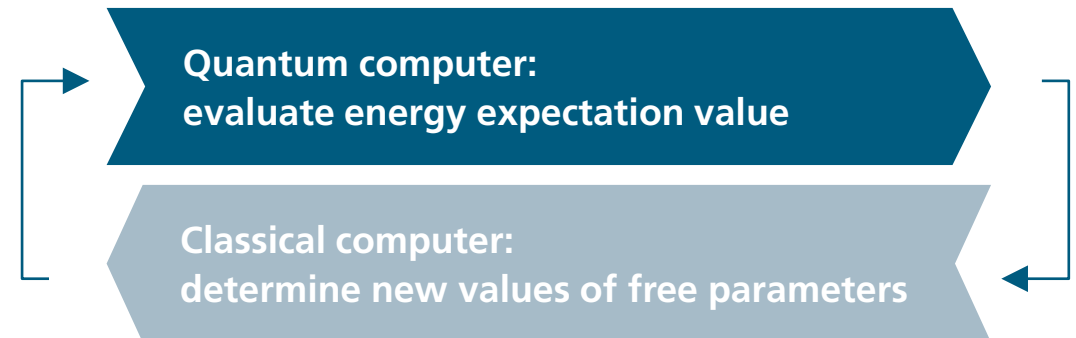
calculate energy expectation value
 $\langle E \rangle$
 for variational state $|\psi(\phi_1, \phi_2, \dots)\rangle$

- Choose ansatz for the preparation of the variational state

$$|\psi(\phi)\rangle = U(\phi)|\psi_{\text{init}}\rangle$$

$$U(\phi) = U_{\text{UCCSD}}(\phi) = \exp \left(\sum_{i\alpha} \phi_i^\alpha a_\alpha^\dagger a_i + \sum_{ij\alpha\beta} \phi_{ij}^{\alpha\beta} a_\alpha^\dagger a_\beta^\dagger a_i a_j - \text{h.c.} \right)$$

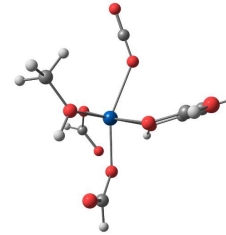
Hybrid quantum-classical algorithm:



Automated Quantum-Chemistry Calculations

Integrated software package

MO	w()	w(↑)	w(↓)	w(⊗)	S	sel
68	0.000	0.003	0.002	0.995	0.037	
69	0.000	0.003	0.001	0.995	0.034	
70	0.000	0.004	0.003	0.993	0.047	
71	0.001	0.003	0.003	0.994	0.045	
72	0.001	0.015	0.002	0.983	0.093	
73	0.005	0.012	0.009	0.974	0.146	
74	0.009	0.006	0.005	0.980	0.119	
75	0.007	0.010	0.008	0.976	0.139	
76	0.003	0.016	0.008	0.974	0.146	
77	0.040	0.003	0.002	0.955	0.203	
78	0.003	0.985	0.000	0.012	0.087	a
79	0.010	0.985	0.001	0.004	0.089	a
80	0.013	0.972	0.004	0.010	0.157	a
81	0.949	0.008	0.008	0.036	0.244	
82	0.971	0.021	0.004	0.003	0.150	
83	0.981	0.003	0.003	0.013	0.110	
84	0.983	0.003	0.003	0.012	0.100	
85	0.976	0.010	0.008	0.006	0.139	
86	0.985	0.012	0.002	0.001	0.087	
87	0.997	0.002	0.001	0.000	0.020	
88	0.998	0.001	0.001	0.000	0.013	
89	0.994	0.003	0.002	0.000	0.040	
90	0.994	0.003	0.002	0.000	0.041	
91	0.996	0.002	0.002	0.000	0.027	



qc_on_qc

Active Space Finder

IBM Qiskit

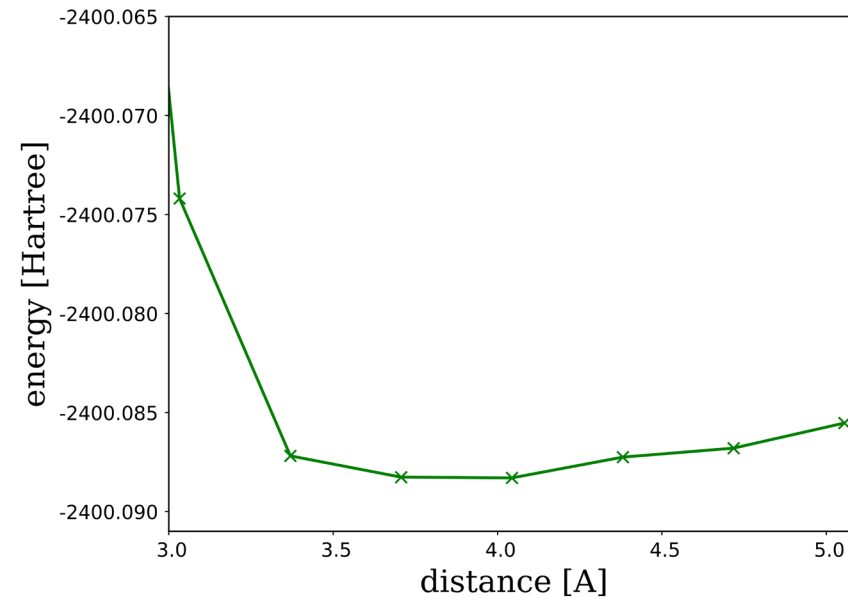
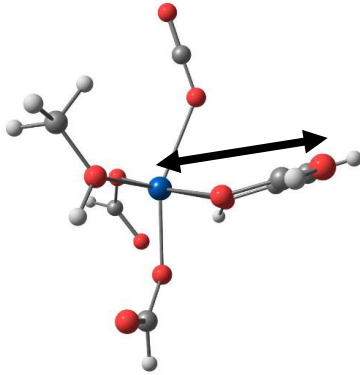
$$\langle E \rangle_{\min}$$

determine the molecular orbitals
with several classical methods and
select the relevant active space

perform VQE calculations

- different components of VQE
- different variants of VQE
 - ↳ different ansätze
 - ↳ AdaptVQE
- including Error Mitigation
 - ↳ readout error mitigation
 - ↳ zero-noise extrapolation

Example of results



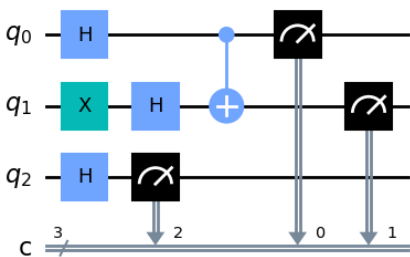
The minimum corresponds to the equilibrium position.

Manuscript in preparation together with open-access publication of software code

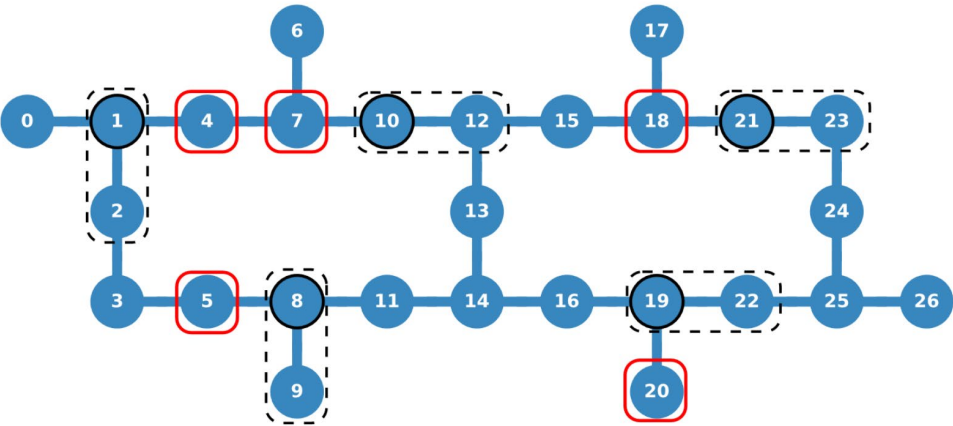
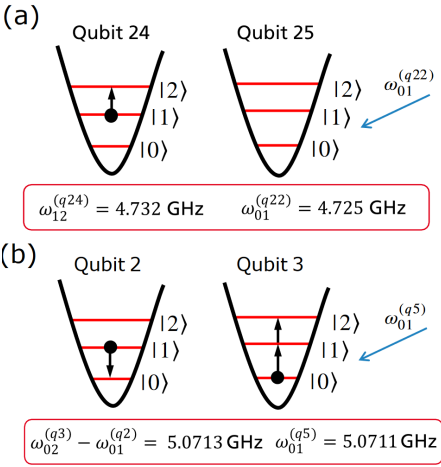
Characterizing crosstalk of superconducting transmon processors

A. Ketterer and Th. Wellens, Phys. Rev. Applied 20, 034065 (2023)

- Ideal picture of quantum circuits:



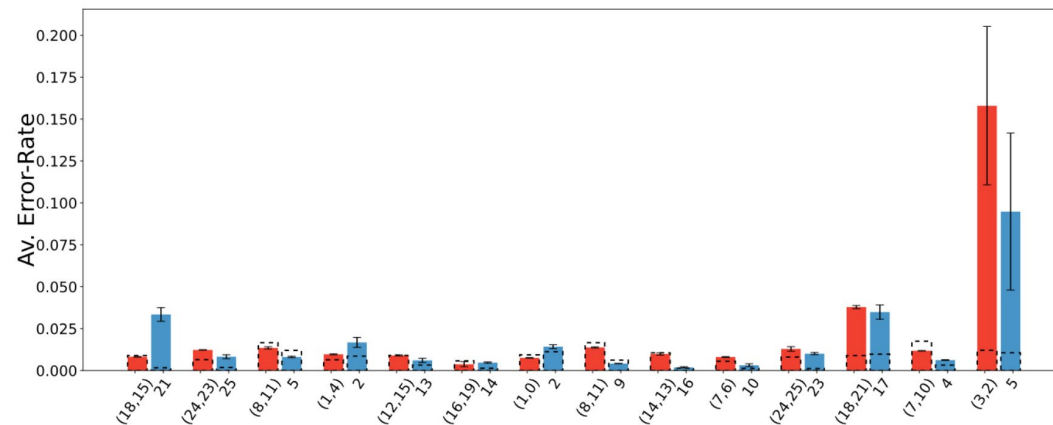
- Correlations between quantum gates on neighbouring qubits
- Correlations are caused by accidental resonances



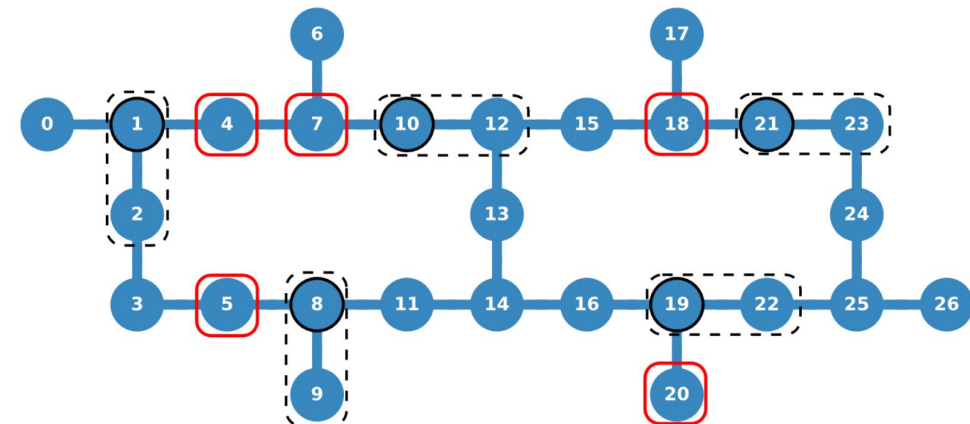
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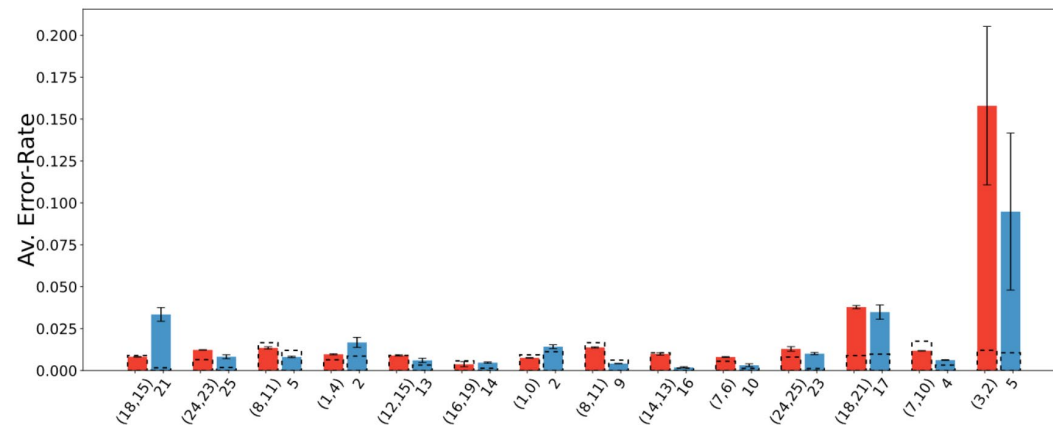
Average error rates for different qubit sets for single- (blue) and two-qubit (red) gates. Dashed boxes indicate the respective average error rates provided by IBM on the same day.



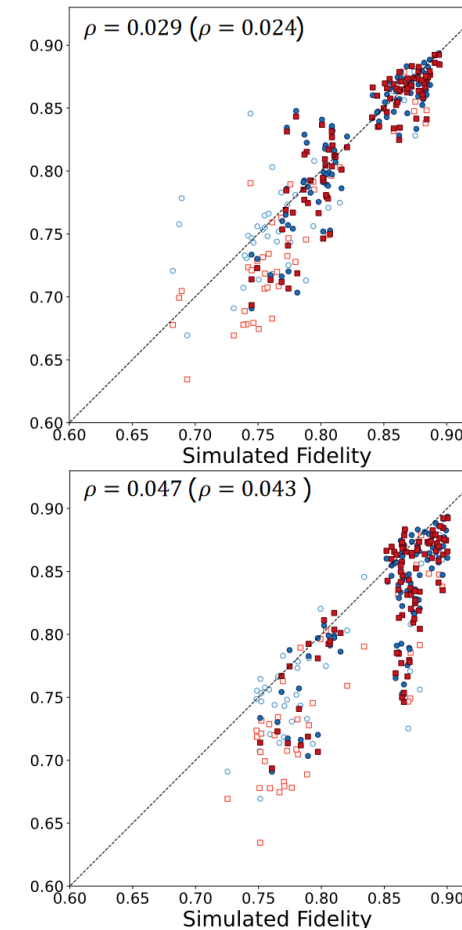
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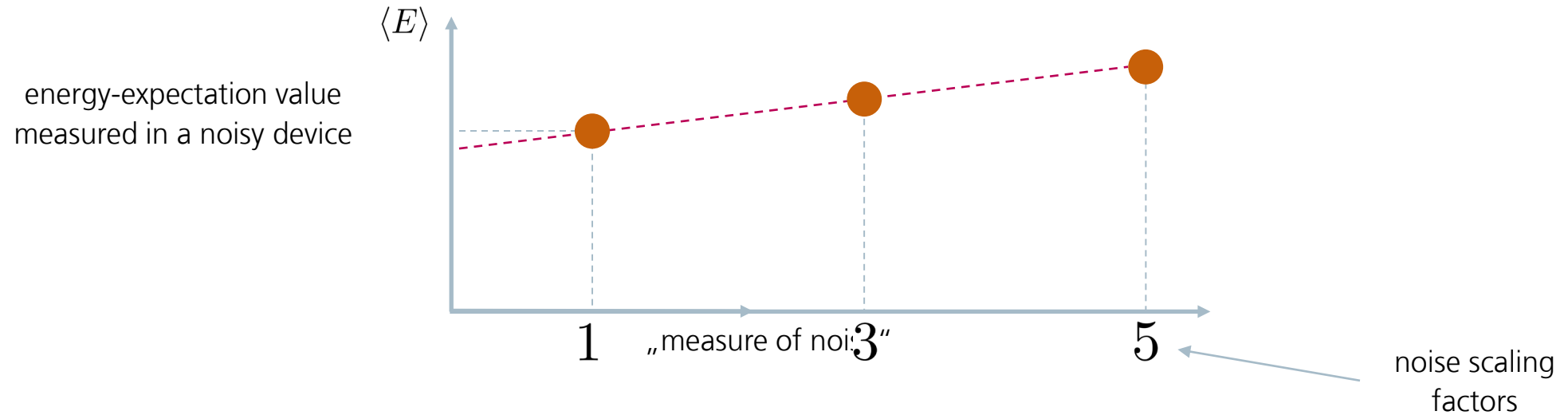


crosstalk-aware noise model

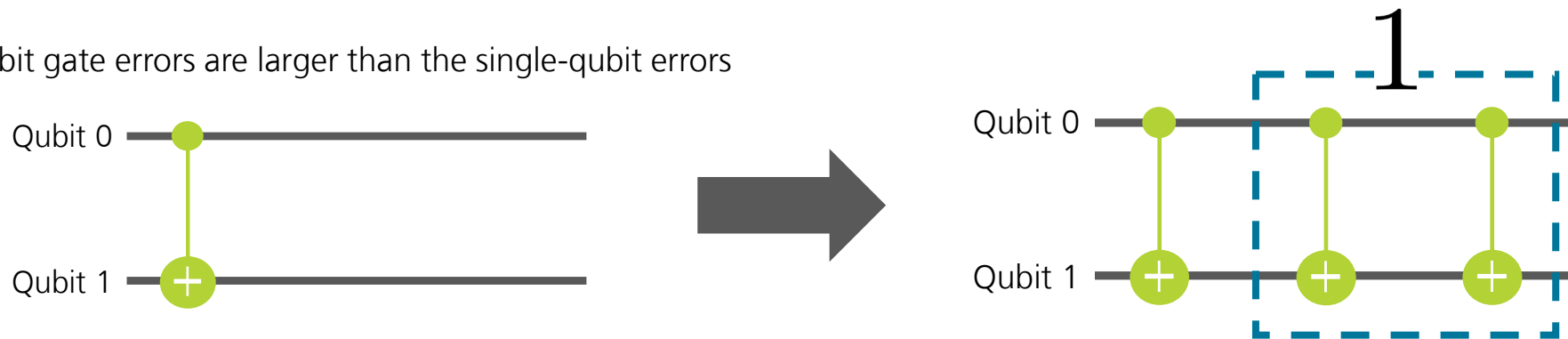
standard IBM noise model

Inverted-circuit zero noise extrapolation

K. König, F. Reinecke and Th. Wellens, PRA 110, 042625



typically, two-qubit gate errors are larger than the single-qubit errors



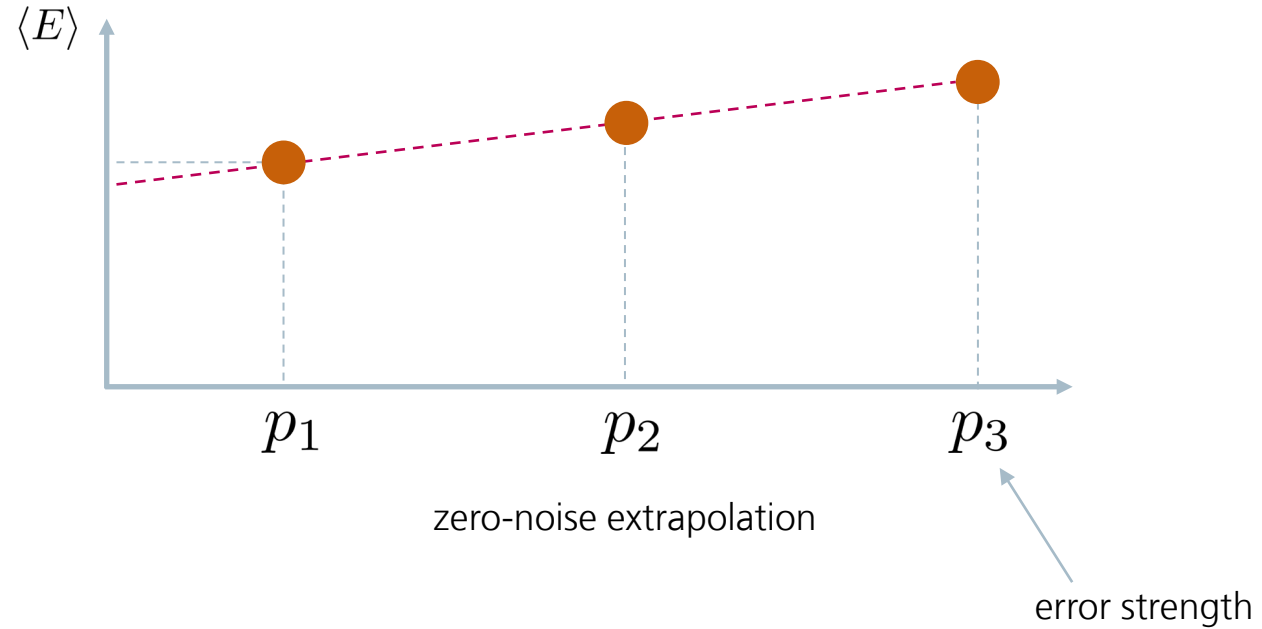
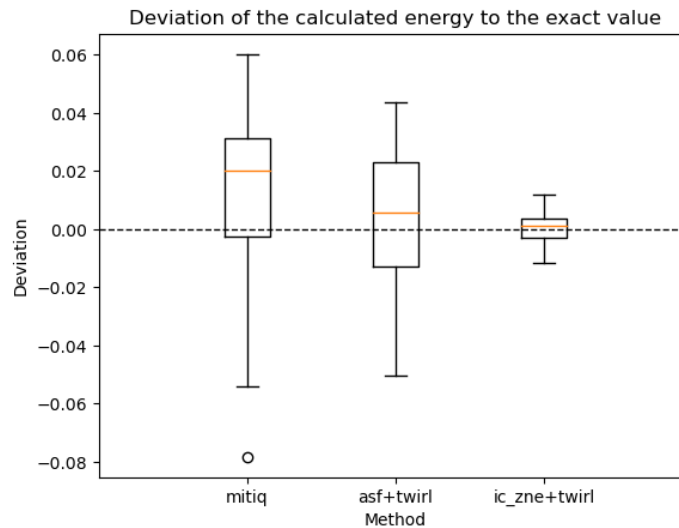
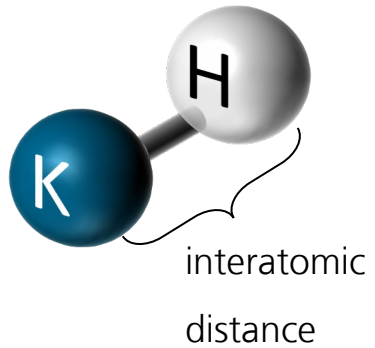
Inverted-circuit zero noise extrapolation

K. König, F. Reinecke and Th. Wellens

better: directly measure the error strength by inverting circuits

$p(P_0)$

probability to successfully
invert the circuit



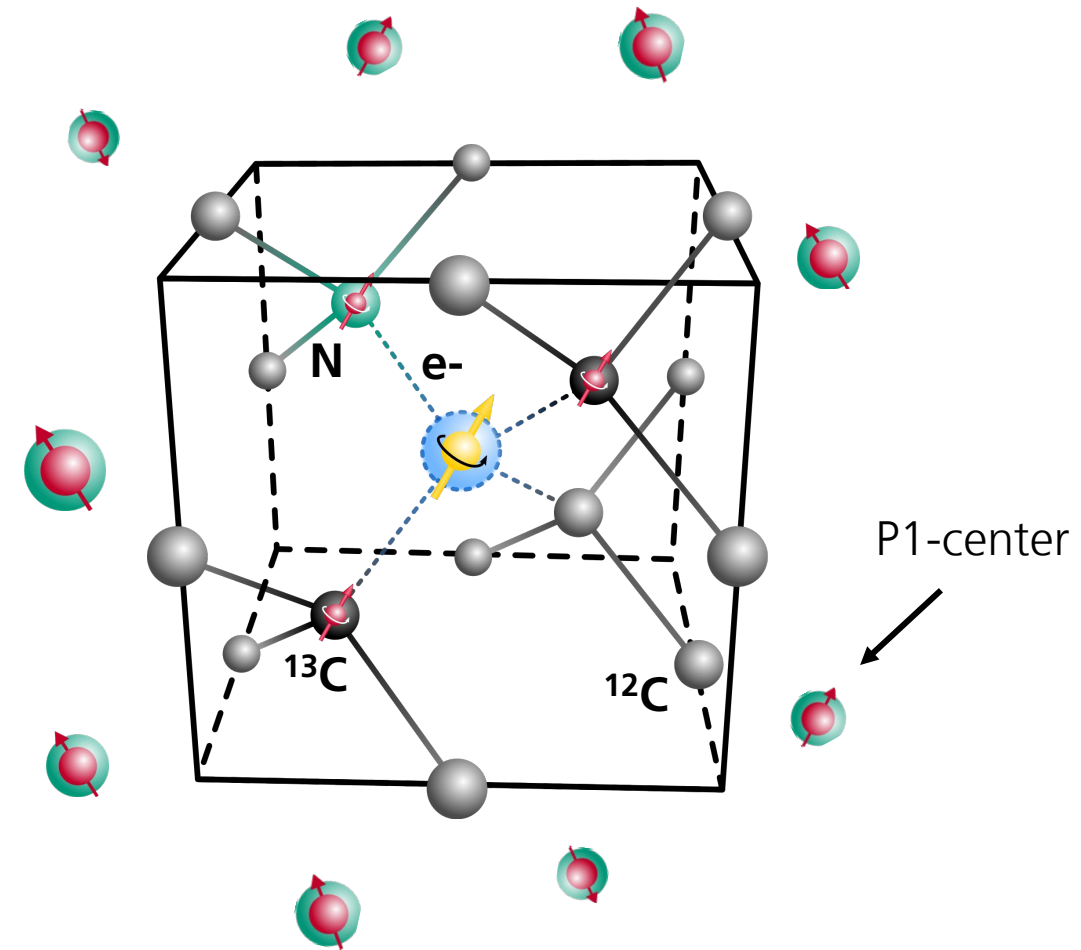
Quantum algorithms research at Fraunhofer IAF

Software-hardware co-design

We pursue various approaches to perform algorithms on the Noisy Intermediate Scale Quantum (NISQ) systems available today, despite their inherent susceptibility to errors.

Methods and algorithms:

- Error characterization
- Error mitigation: Zero Noise Extrapolation (IC-ZNE)
- **Simulation of spin models** →
 - Additional defects affect the coherence properties of the NV center
 - For example, single-substitutional nitrogen defect (P1-centers) form a strongly-coupled spin bath → shorten the coherence time



P. Schätzle et al., Spin coherence in strongly coupled spin baths in quasi-two-dimensional layers, Phys. Rev. B 110, L220302 (2024)

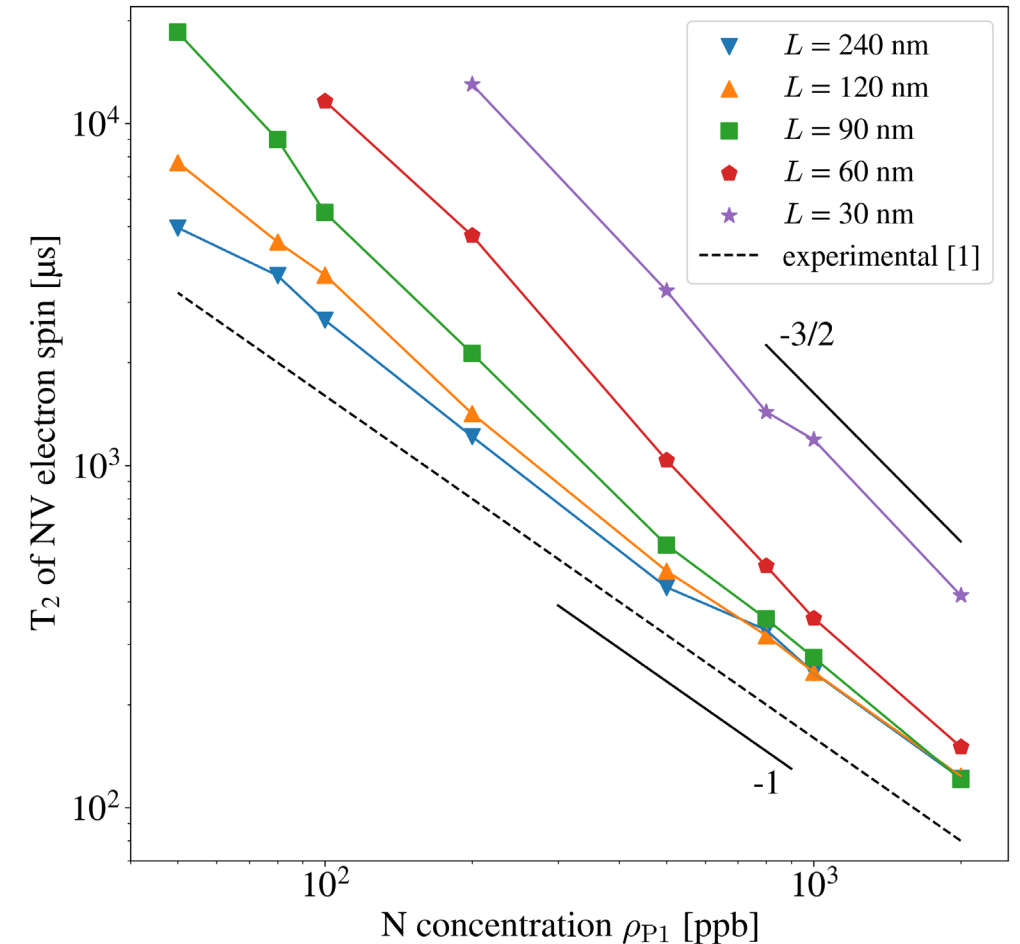
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- **Simulation of spin models** →
 - Additional defects affect the coherence properties of the NV center
 - For example, single-substitutional nitrogen defect (P1-centers) form a strongly-coupled spin bath → shorten the coherence time
 - We introduced **partition coupled-cluster expansion (pCCE)** to group defects by interaction strength and capture correlations
 - *Our results are more accurate than previous (Gauss-Markov) models*



P. Schätzle et al., Spin coherence in strongly coupled spin baths in quasi-two-dimensional layers, Phys. Rev. B **110**, L220302 (2024)

Quantum computing with color centers in diamond

Quantum optimization

Quantum optimization

Dehn et al., arXiv:2504.08577

- “Quantum Approximate Optimization Algorithm” (QAOA)

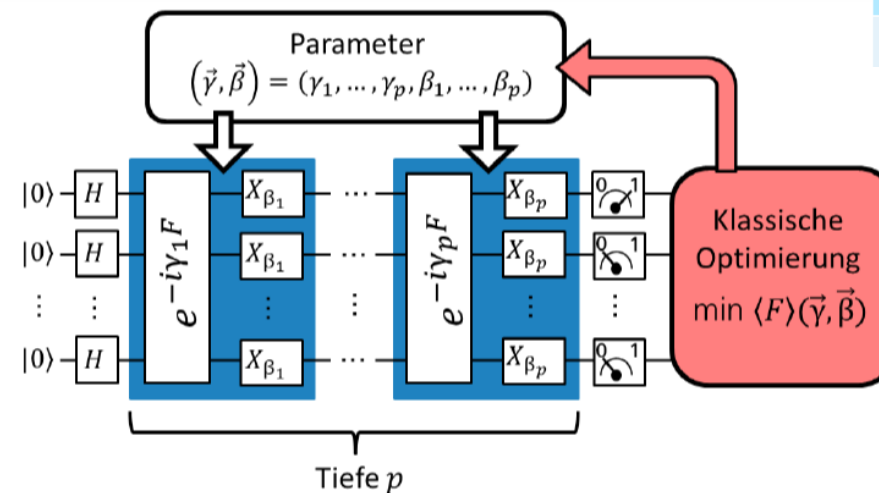
- Formulation of a combinatorial optimization problem through a binary function to be minimized (QUBO)

$$F(z_1, \dots, z_n) = \sum_{i,j=1}^n F_{ij} z_i z_j$$

- Optimization through a parametrized quantum circuit → **hybrid algorithm**
- Measurement: for each qubit result 0 or 1, thus $z_i=0$ or 1 for qubit $i=1, \dots, n$, hence a candidate for a potential solution of the optimization problem

asset	choice z_i 1: yes, 0: no
adidas	1
Allianz	0
BASF	0
Bayer	1
Beiersdorf	1
BMW	0

QAOA circuit for portfolio optimization

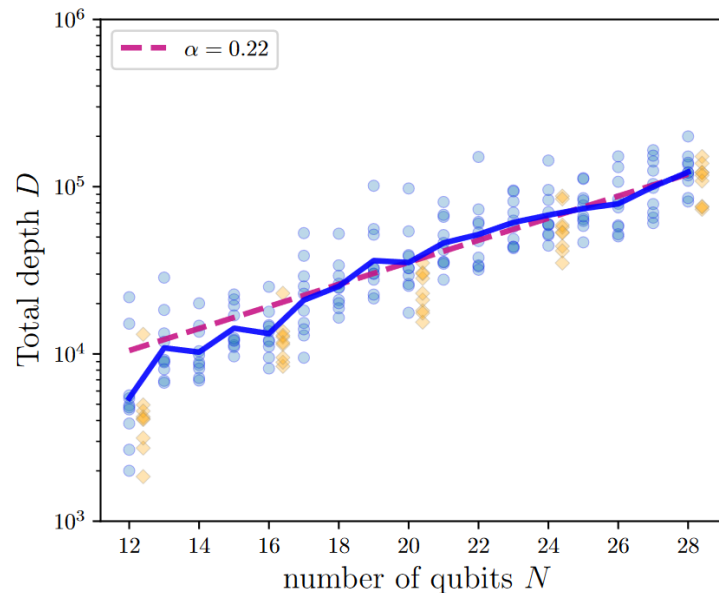


Applications:

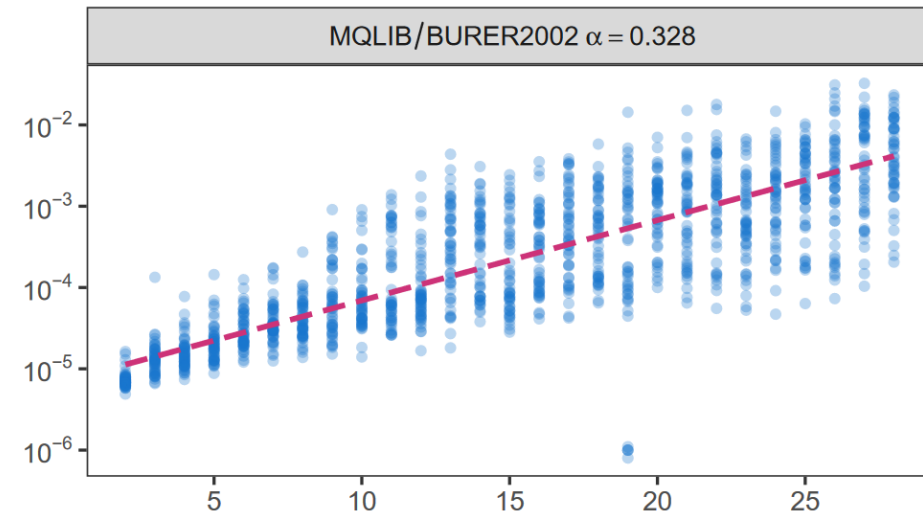
Portfolio analysis in collaboration with industry partner as use-case provider
Ressource management, assignment, and decision problems
Advantage compared to classical optimization routines?

Results on a potential scaling advantage in quantum optimization

Quantum calculation



Classical calculation



- Simulation of up to 28 qubits → Change of the scaling behavior observable
- Result still ambiguous → more qubits (of good quality) or larger simulation necessary

Quantum computing at Fraunhofer IAF

Conclusion and outlook

Conclusion and outlook

Quantum computing at Fraunhofer IAF

Support and pioneer diamond quantum technologies (quantum computing and sensing)

- We build and operate **quantum computers**, especially with **color centers in diamond**
- Material development and qubit creation in diamond: **nitrogen-vacancy (NV)** centers
- **Pushing the scalability frontier:** Dipolar / photonic coupling architectures
- **Future:** single-ion implantation, other types of color centers
- **Quantum algorithms research:**

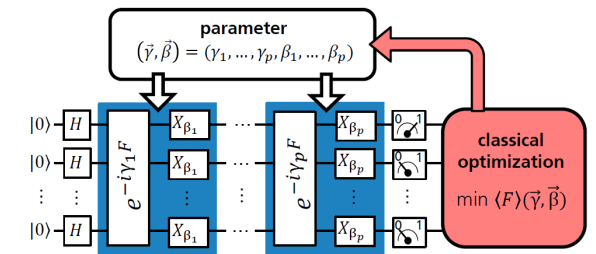
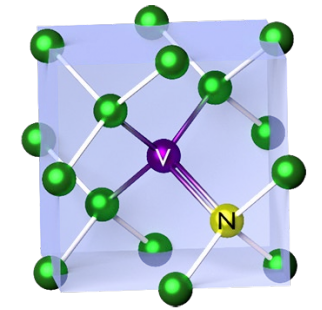
Qubit and noise characterization, error mitigation

Quantum simulation and optimization and machine learning

Outlook

HPC integration of NV quantum computing platforms

Implementation of quantum algorithms on NV platforms



Contact

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